

Applying Machine Learning Algorithms to Stock Price Forecasting and Causal Analysis of ESG Effects on Returns

Cong Doan Truong²²⁰⁷⁰⁷¹⁵, Le Ha Trang²²⁰⁷⁰⁷¹⁵, Luong Vu Hoang Anh²²⁰⁷⁰⁶⁷³, Bui Thi Diem Quynh²²⁰⁷⁰⁶²⁶, Le Minh Hien²²⁰⁷⁰⁷⁰⁹, Doan Huong Giang²²⁰⁷⁰⁷²⁵
International School, Vietnam National University Ha Noi
tcdoan@vnu.edu.vn
22070715@vnu.edu.vn
22070673@vnu.edu.vn
22070626@vnu.edu.vn
22070709@vnu.edu.vn
22070725@vnu.edu.vn

Abstract. ESG has become a strategic factor in driving sustainable value and long-term investment decisions, while also raising the question of whether these requirements contribute to the financial performance of businesses. Therefore, this study analyzes the integration of ESG scores and ESG risk measures into stock price forecasting models to examine how ESG factors influence daily adjusted closing prices. The dataset collected 296,334 observations from 303 publicly listed stocks between February 14, 2019, and December 30, 2022. We mix ESG scores with basic company factors and price-based characteristics to evaluate the performance of various machine learning models. The results showed linear models had the most stable predictive performance, with linear regression reaching $R^2 = 0.999023$ and $MAE = 2.095274$ on the validated set, outperforming tree-based models with $R^2 \approx 0.97$ and $MAE \approx 6.2\text{--}6.4$. SHAP shows that short-term price changes, particularly recent price lags and moving averages, dominated daily price forecasts, and ESG variables contributed only a little increase in predictive power. To address the endogeneity issue, double machine learning was used to examine the causal impact of ESG on corporate profitability. The findings showed heterogeneous effects among the ESG risk components, such as the tendency for profitability to offset environmental risk, the impact of social risk on performance, and the small effects of governance. However, ESG scores are associated with positive effects, which become significant over longer periods. In general, the findings suggest that ESG indicators are not very suitable for short-term stock price forecasting models, but they have a significantly stronger impact on long-term average returns. Future work will expand the analysis to richer ESG datasets and shift the model focus from price prediction to return- and risk-based targets to more directly assess the impact of ESG on valuation and investment decisions.

Keywords: ESG-aware valuation; financial modeling; ESG score; ESG risk; Stock price prediction; SHAP; Machine learning; Double Machine Learning (DML); endogeneity; time-series validation.

1 Introduction

In recent years, environmental, social, and governance (ESG) frameworks have increasingly become important assessment tools for assessing the sustainability and responsibility of companies facing environmental and social issues [1]. With growing concerns about globalization, climate change, human rights, and corporate social responsibility, ESG criteria are no longer viewed as short-term or one-time initiatives. Instead, it is increasingly regarded as a strategic instrument integrated into financial and investment decisions, contributing to long-term value creation in global financial markets. In practice, however, the implementation of sustainability-oriented policies typically requires substantial financial resources, raising a critical question as to whether ESG scores and ratings genuinely contribute to firms' financial performance. This question has attracted considerable attention not only from corporate managers and investors but also from the academic community. Studies highlight the importance of ESG in mitigating risk, stabilizing stock prices, and enhancing business resilience during periods of market volatility. Evidence from the COVID-19 era shows that companies with high ESG scores and ESG-related stocks demonstrated rapid and robust recovery [2], [3]. This finding reinforces the view that ESG functions as a risk-mitigating shield during times of instability. Furthermore, differences in ESG ratings significantly increase the risk of a stock price crash by undermining investor confidence, while prices gradually adjust to changes in ESG ratings, reflecting the long-term importance of ESG value in its interaction with governance and market conditions [4]. Another approach argues that ESG does not affect the stock market as a whole. This study shows that the environmental pillar (E) plays an important role in the transmission of information and influence on short-term stock market returns. This study shows that the environmental pillar (E) plays an important role in the transmission of information and influence on short-term stock market returns [4]. Conversely, the governance pillar demonstrates a strong impact on risk mitigation; this study shows that increasing governance by one point reduces the risk of a stock price decline by 0.39 units [4]. Meanwhile, the social pillar (S) has inconsistent influence across markets and is only meaningful in specific circumstances, for example, where investors prioritize community involvement, labor standards, and employee relations [5]. However, some studies suggest that ESG primarily reflects social norms rather than financial values. Evidence indicates that ESG lacks the ability to predict stock returns and does not have a direct causal relationship with short-term stock price downside risk in the market [6], [7]. Although research on ESG criteria and financial performance is rapidly developing, there are still gaps in assessing the predictive value of ESG criteria in machine learning, as well as in assessing the relative impact of each ESG pillar on forecast accuracy. A significant limitation is the lack of a unified ESG standard, which leads to inconsistent model inputs depending on the rating provider. Existing models often struggle to detect long-term dependencies and lack the need for systematic hyperparameter optimization, which limits forecast performance [8]. Furthermore, commonly used algorithms such as decision trees are prone to overfitting of training data, which reduces their ability to generalize to unobserved observations [9]. To address these gaps, this study examines the role of ESG factors in machine

learning models predicting stock prices and applies causal inference to analyze the impact of ESG factors on average monthly and annual returns.

2 Methodology

2.1 Project workflow

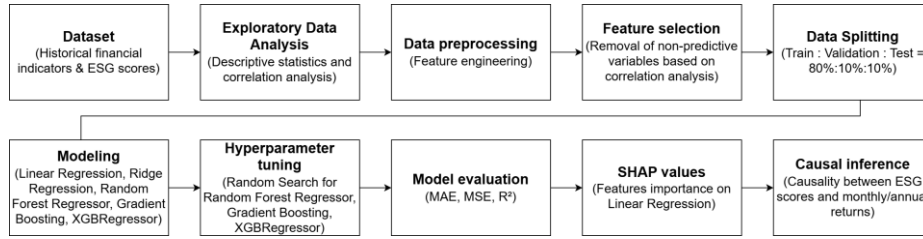


Fig 1. Overview of the proposed research pipeline

Figure 1 presents the overall methodological framework of the study. The study was programmed in Python and executed on the Google Colab platform. Data were preprocessed, modeled, and evaluated using multiple regression and machine learning techniques. SHAP-based interpretation and causal inference were subsequently applied to analyze feature importance and the causal relationship between ESG scores and returns.

2.2 Data

The empirical analysis is based on a daily dataset covering 303 publicly listed equities over the period 2019-02-14 to 2022-12-30, yielding a total of 296,334 firm–day observations. Historical financial and price data were collected from Yahoo Finance (yfinance), while ESG-related data were obtained from other external ESG data sources. Each observation corresponds to a firm (identified by ticker and name) on a specific trading date. The dataset contains equity price information in the form of adjusted closing prices used for price prediction and return construction, firm-level financial characteristics and price-derived technical variables, and pillar-level ESG information comprising Environmental, Social, and Governance scores together with the corresponding ESG Risk Scores.

For preprocessing, the dataset is first checked for missing values and ordered chronologically to preserve the temporal structure. To reduce dimensionality, correlations among numerical variables are examined and highly correlated features are removed. Then, we calculate `lag_market_cap` to mitigate scale imbalances across variables. The processed data are then split into training, validation, and test sets using an 80:10:10 ratio based on temporal ordering.

2.3 Models

To have a robust evaluation of prediction performance, five popular modeling approaches including Linear Regression, Ridge Regression, Random Forest (RF), Gradient Boost and XGBoost are applied in this research. These range from more simplistic to complex models, enabling a systematic comparison of interpretable linear methods and flexible nonlinear aggregation approaches as recommended by best practices in predictive modeling [10].

Linear Regression

The study relies on linear regression primarily because of its interpretability, computational efficiency and theoretical roots in statistical learning. The model assumes a linear association between the predictor and response variables such that the estimated coefficients are interpreted as boundaries directly. Although fairly restrictive Linear Regression is a standard benchmark in empirical work and so makes for a good reference point to measure improvements from employing more complex statistical/machine learning models [10].

Ridge Regression

Ordinary Linear Regression is straightforward and interpretable, but it tends to suffer from multicollinearity problems and overfitting if the X features have a very large number of high correlated candidates. To address these issues, this work uses Ridge Regression, an adjusted extension of the linear model. The L2 penalty on the coefficient size of a linear model shrinks parameter estimates and reduces variance, aiding out of sample generalization. This adjustment is especially useful since it stabilizes the linearity while increasing predictive accuracy in the presence of multicollinearity [11].

Random Forest

In order to detect nonlinear interactions which linear models will not consider, this study includes Random Forest, a strongly tree-based ensemble learning method. This algorithm creates multiple decision trees, trains each on an initialized sample of the data, and then average its predictions which in turn minimizes model variance and makes overfitting less likely. Random Forests is the gold standard in applied predictive modeling because it is capable of dealing with complex data structures and high-dimensional feature spaces without strong parametric assumptions [12].

Gradient Boosting

Gradient Boosting is a propaedeutic aggregate learning approach, every new tree tries to eliminate the remaining error of the learnt model while being trained from its predecessors. Such iterative optimization procedure enables to learn subtle nonlinear relationships and interactions among the variables in order to minimize a prescribed loss function. In numerous real-world applications, the proposed framework has exhibited excellent predictive power, particularly within structured table data where learning high-order dependencies over a sequence is key [13].

XGBoost

XGBoost (Extreme Gradient Boosting) is an implementation of gradient boosting on the decision trees to work more efficiently and faster. This algorithm has advanced tuning capabilities to avoid over fitting and good performance for handling missing values, parallel computation that improves prediction accuracies as well as system

speed. Thanks to these enhancements, XGBoost has grown into one of the go-to models in applied machine learning, particularly for complex and high-dimensional data [14].

2.4 Model evaluation

Mean Absolute Error (MAE) measures the average absolute difference between predicted values and actual outcomes, without considering the direction of the errors.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

Mean Squared Error (MSE) measures the average of the squared differences between the predicted values and the actual observations. MSE is always non-negative, and values closer to zero indicate better model performance.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

R-squared (R^2), also known as the coefficient of determination, is a statistical measure that the extent to which variability in the dependent variable can be explained by the set of independent variables.

$$R^2 = 1 - \frac{RSS}{TSS} \quad (3)$$

2.5 SHAP Value

The SHAP (Shapley Additive exPlanations) method is used to explain the decision-making mechanisms of machine learning models and to assess the contribution of each input variable to the prediction results. SHAP was chosen because of its ability to provide consistent explanatory values, its strong theoretical basis, and its applicability to a wide range of models [15]. Classic Shapley Value Estimation:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)] \quad (4)$$

Where:

- F : set of all features
- S : a subset that does not contain i
- $[f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)]$: predicted change level when adding feature i to set S

2.6 Causal Inference by using Linear Double Machine Learning

Double-Debiased Machine Learning, or DML, is a statistical and theoretical framework to fine the causal inference affect. It can estimate the causal effect of a treatment variable T on an outcome variable Y , while accounting for multiple confounding variables X , by leveraging the powerful predictive capabilities of machine learning [16]. DML is built upon a Partial Linear Regression (PLR) model:

$$Y = T\theta_0 + g_0(X) + \zeta, E(\zeta|D, X) = 0 \quad (6)$$

$$T = m_0(X) + V, E(V|X) = 0 \quad (7)$$

Where:

- Y : output, T : Treatment variable, X : Confounder variables
- θ_0 : Causal parameter
- $g_0(X)$: Outcome model $E[Y|X]$, $m_0(X)$: Treatment model $E[T|X]$
- ζ, V : error terms

DML performs causal inference in three steps. First, it uses machine learning models to estimate the relationship between X and Y , $g_0(X)$, and the relationship between X and D - $m_0(X)$. Next, DML removes the influence of confounding variables by taking the residuals between the actual and predicted values. Finally, the model estimates the causal parameter through OLS regression between the residuals to find causality θ_0 . By exploiting Neyman orthogonality and sample splitting with cross-fitting, DML ensures that estimation errors in the nuisance functions do not bias the causal effect, enabling stable and unbiased causal inference in high-dimensional settings [16].

3 Results and discussion

3.1 Model performance evaluation

Table 1. Model performance evaluation across training, validation, and test sets

Model	MAE	MSE	R^2
Training Performance Results			
Random Forest	4.1818	573.6319	0.9710
Gradient Boosting	4.0468	540.7457	0.9727
Linear Regression	1.5030	13.3846	0.9993
Ridge Regression	1.5030	13.3848	0.9993
XGBoost	10.5399	4323.1597	0.7814
Valid Performance Results			
Random Forest	6.4440	683.6215	0.9694
Gradient Boosting	6.2244	634.2495	0.9716
Linear Regression	2.0953	21.8223	0.9990
Ridge Regression	2.0955	21.8230	0.9990
XGBoost	20.4023	11110.0862	0.5027

Test Performance Results			
Random Forest	6.2724	583.3503	0.9718
Gradient Boosting	6.0936	528.9266	0.9745
Linear Regression	1.9300	15.4596	0.9993
Ridge Regression	1.9306	15.4623	0.9993
XGBoost	19.2099	10781.2085	0.4794

Linear and Ridge Regression perform as the top two models, with an optimal prediction accuracy. For both models below are the low MAE and MSE while retaining R^2 values of ~ 0.99 , indicating that it maintains a remarkable stability over datasets with no indication of overfitting. Random Forest and Gradient Boosting shape a second performance tier, showcasing decent R^2 values in the range of (0.96-0.97) but are too flexible to obtain marginal improvements over linear benchmarks in our use case. On the other hand, XGBoost fares worse, and its larger error measures point to its more elaborate regularization not being a good fit for the data's true shape.

These differences come from basic design variations. Methods using tree structures depend on dividing data into groups based on a few very important factors, making them easily affected by changes in those factors [12]. On the other hand, linear methods spread data across all input factors by using weighted combinations, leading to more stable predictions [10]. As a result, even though combined models are useful for checking stability, linear methods are still the best option for this situation. Here is the demo web of models: Stock Prediction Demo - a Hugging Face Space by changle-2110.

3.2 SHAP value

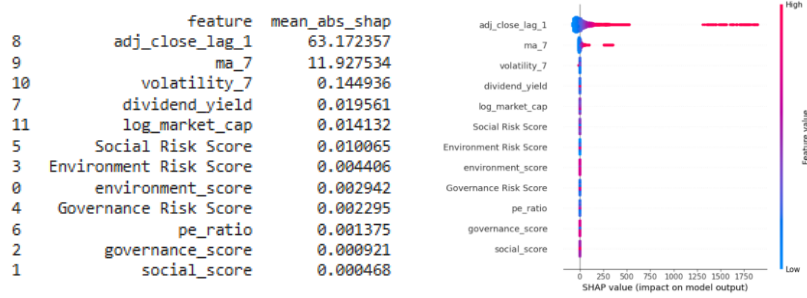


Fig 2. SHAP feature importance for the stock price prediction model

Figure 2 shows that the forecasting model relies primarily on past price information. Based on the results, the two variables with the most influence on the model are `adj_close_lag_1` (with a result of ≈ 63.17) and `ma_7` with `mean_abs_shap` of ≈ 11.93 . Higher values for these variables generally increase the predicted stock price. This demonstrates a positive relationship between past prices and the model's forecast. Meanwhile, other fundamental financial variables have little impact on the forecast results. Specifically, the mean SHAP values of `volatility_7` (≈ 0.145), `dividend_yield` (≈ 0.0196), `log_market_cap` (≈ 0.0141), and `pe_ratio` (≈ 0.00138) are close to zero, indicating negligible contributions.

The scores for environmental, social, and governance factors, as well as the overall ESG risk score, all have mean_abs_shap below 0.01. This is evident in the short-term stock price forecasting, where the model prioritizes time-series signals based on price volatility over ESG characteristics. In short, ESG factors have not proven effective in predicting daily stock prices within the framework of the analyzed model.

3.3 Casual inference

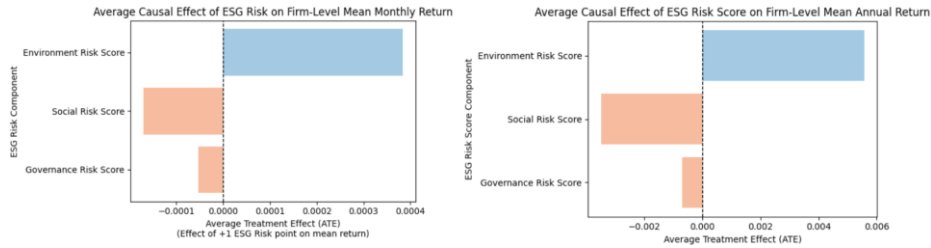


Fig 3. Average causal effect of ESG risk score on firm-level mean monthly/annual

Figure 3 illustrates the consistent impact of ESG risk components on business profitability at different time levels. Environmental risk scores consistently show a positive impact on monthly and annual profits. This confirms that high environmental risk scores contribute to increased average business profitability. Conversely, social risk has a negative impact. This also shows that social issues negatively affect business performance in both the short and long term. Governance risk tends to be negative, but to an insignificant degree. These results suggest that ESG risk scores do not contribute to the overall financial performance of the business as a whole. Each pillar creates a distinct impact, where environmental risk is offset by profitability, while social and governance risks directly harm business performance.

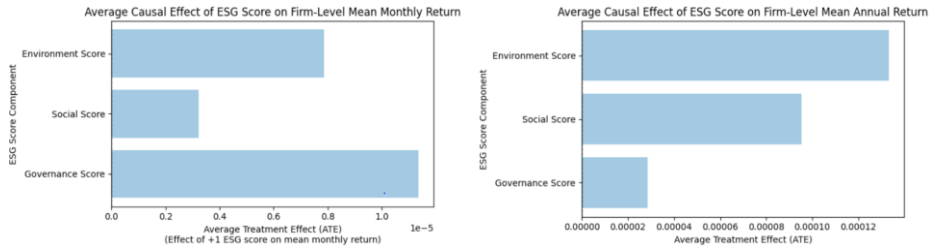


Fig 4. Average causal effect of ESG score on firm-level mean monthly/annual return

Figure 4 illustrates the causal relationship between ESG score and firm-mean return in both monthly and annual periods. The graph shows that all estimated impacts are positive. This demonstrates that improvements in scores across all three ESG pillars lead to a direct increase in the firm's mean return. In the monthly period, the governance score shows the strongest influence. This is logical because improvements in governance, such as increased shareholder power, are quickly valued by the market. However, in

the long term, the environmental and social scores show a larger and more pronounced impact on mean return. This suggests that green strategies and corporate social responsibility initiatives take time to translate into tangible economic value, consistent with the cumulative and long-term impact of ESG.

4 Conclusion

This study shows that ESG indicators are not effective tools for forecasting daily stock prices; instead, technical variables such as one-day lagging prices and moving averages dominate in daily forecasting models. Linear models such as Linear Regression achieve very high accuracy with $R^2 = 0.999023$ and rely primarily on historical price data. However, ESG does impact long-term average returns, such as monthly and annual returns. Through causal analysis using Dual Machine Learning (DML), the study indicates that environmental risk can still lead to higher returns, while social risk directly and negatively impacts stock returns. Conversely, improving ESG scores yields positive results. The governance score has the strongest impact on monthly returns. Meanwhile, the remaining factors have a stronger influence on long-term growth. In the future, we will expand our research with newer and richer ESG data. At the same time, instead of focusing solely on short-term data, we will shift our research from price forecasting to pricing frameworks based on return and risk.

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